UNIT 3

1. Data Transformation:

Data Transformation is the process of converting raw data into a clean, structured, and analysis-ready format. In tools like Power BI, Power Query, Excel, or SQL, transformation ensures that the data is accurate, consistent, and useful for analysis or reporting.

Data Transformation is the process of modifying, cleaning, structuring, and enriching raw data into a suitable format for analysis and visualization.

Data Transformation is a key step in ETL (Extract → Transform → Load).

In Power BI, data transformation is mainly done using Power Query Editor, which allows users to perform ETL (Extract, Transform, Load) operations.

Data Transformation is Important

- Removes errors, duplicates, and inconsistencies.
- Converts data into a standard format.
- Improves data quality and reliability.
- Helps in building dashboards, reports, and predictive models.
- Ensures data follows business rules.

Steps in Data Transformation

a. Data Cleaning

- Handling missing values (replace/remove)
- Removing duplicates
- Correcting spelling errors
- Standardizing formats (date, numbers, text)

b. Data Structuring

- Adding new columns
- Splitting/merging columns
- Changing data types
- Creating calculated fields

c. Data Integration

- Combining data from multiple sources
- Merging or appending tables

d. Data Reduction

- Filtering rows
- Removing unwanted columns
- Aggregating (sum, average, group-by)

2. Power Query Editor:

Power Query is the data transformation engine inside Power BI. Power Query Editor is a data transformation and preparation tool available in **Power BI**, **Excel**, and **Power Query Online**.It allows users to **connect, clean, reshape, and prepare** data before loading it into the data model.Power Query works using a backend formula language called **M Language** (Mashup Language). Each transformation you apply becomes a **step** in the "Applied Steps" pane.

Features of Power Query

Geetanjali

- Import from multiple sources
- Automatically records steps
- User-friendly UI
- Supports formulas (M-language)
- Fast processing for large datasets

Power Query Editor is a **no-code**/low-code interface in Power BI Desktop where you:

- **Connect to data sources**
- **Clean and transform data**
- **Shape data before loading it into the data model**

To open it.

Home Tab → Transform Data → Power Query Editor

Use of Power Query Editor

- To clean and prepare data for analysis
- To automate repetitive cleaning tasks
- To merge data from multiple sources
- To maintain data accuracy and consistency
- To transform large datasets efficiently

Power BI dashboards heavily depend on correct transformations performed in Power Query.

Interface Overview

I. Ribbon Tabs

Power Query Editor contains several tabs:

a. Home Tab

Most commonly used operations:

- Remove columns/rows
- Keep rows
- Merge queries
- Append queries
- Use first row as header
- Close & apply

b. Transform Tab

Used to modify data WITHOUT adding new column:

- Data type
- Pivot / Unpivot columns
- Replace values
- Reverse rows
- Split columns
- Format text

c. Add Column Tab

Used to add new columns:

- Custom column
- Conditional column
- Index column

geetanjali

- Duplicate column
- Extract values

d. View Tab

Used to control visibility:

- Formula bar
- Column quality
- Column distribution
- Advanced editor

Key Components of Power Query Editor Interface

I. Query Pane

Left side — shows all tables/queries you have loaded.

II. Data Preview Pane

Displays data after each transformation.

III. Applied Steps Pane

Right side — shows each transformation step.

You can:

- Rename steps
- Delete steps
- Reorder steps
- View M code for each step

Connecting and Importing Data

Power Query supports many data sources:

- Excel, CSV
- SQL Server, MySQL
- Web data
- SharePoint
- PDF
- Folder (for combining multiple files)

Steps

Home → New Source → Choose file → Open in Power Query Editor

Most Important Data Transformation Functions

Use First Row as Headers

Power Query uses "Column1, Column2" by default.

Steps

Home → *Use First Row as Headers*

Why?

Meaningful column names improve readability and accuracy.

Remove Rows

Used to eliminate unnecessary rows.

Options

- Remove Top Rows
- Remove Bottom Rows
- Remove Alternate Rows
- Remove Blank Rows
- Remove Duplicates

Usage

Geetanjali

Removing:

- Subtotals
- Summary lines
- Extra headers
- Blank or null rows

Keeping Rows
Opposite of removing rows.
Options
- Keep Top Rows
- Keep Bottom Rows
- Keep Range of Rows

Usage
Keep only required sample data.

Remove Columns
Used to drop irrelevant data.
Examples:
- Address fields not required
- Temporary columns
- Audit columns

Be careful: Removing columns early affects further steps.
Filter Rows
Filtering rows based on:
- Text
- Number
- Date
- Custom conditions

Examples:
- Age > 18
- State = "UP"
- Date between 2022–2023

Change Data Type
One of the most important transformation steps.
Common Types
- Whole Number
- Decimal Number
- Date
- Date/Time
- True/False
- Text

Incorrect data type results in:
- Errors
- Wrong visuals
- Wrong sorting

Split Column

Used when one column contains multiple values.

Geetanjali

Types

- By Delimiter (comma, space, '-')
- By Number of Characters
- By Positions

Examples

- Full Name → First Name, Last Name
- "123-ABC-789" → 3 parts

Merge Columns
Used to combine multiple fields.
Example:
FirstName + LastName = FullName
Replace Values
Used to correct data.
Examples:

- Replace "N/A" with null
- Replace "UP " with "UP"
- Replace misspelled category names

Fill Up / Fill Down
Used when data is missing but available in previous/next row.
Example:
Category listed once for multiple items.
Fill Down copies the above value; Fill Up copies below.
Group By
Used to create summarizations.
Examples

- Total Sales per Region
- Count Orders per Customer
- Average Marks per Student

Remove Errors
Fixes:

- Invalid data types
- Nulls causing errors
- Wrong calculations

Useful for cleaning CSV files.
Pivot & Unpivot Columns
a. Pivot Column
Converts rows into columns.
Example:
Month-wise Sales table.
b. Unpivot Column
Converts multiple columns into attribute-value pairs.
VERY important for making data analysis-friendly.
Example:
Year columns (2020, 2021, 2022) → Converted into:
Year | Value
Conditional Column

Geetanjali

Creates classification columns using IF-ELSE logic.

Example

If Score ≥ 90 → "Excellent", else "Good"

No need for formulas — UI based.

6.15 Custom Column

Used to write formulas using M Language.

Example

New column = [Quantity] * [Price]

Index Column

Creates Row Number:

- From 0
- From 1
- Custom

Useful for sorting or uniquely identifying rows.

Merge Queries

Used to combine columns from two tables.

Join Types

- Inner
- Left Outer
- Right Outer
- Full Outer
- Anti (Left/Right)

Example:

Add department name into student table using DeptID.

6.18 Append Queries

Used to STACK tables vertically.

Example

Sales_2023 + Sales_2024 = Combined Sales Table

3. **Remove Rows**

Used to clean data by removing irrelevant, duplicate, or unwanted rows.

Types of Remove Row Options

a) Remove Top Rows

- Deletes a specific number of rows from the top.
- Useful when:
    - Extra headings
    - Extra blank rows
    - File metadata is present

b) Remove Bottom Rows

- Removes the last few rows (common in system generated reports).

c) Remove Alternate Rows

- Used for repeated pattern data.

Geetanjali

- Example: Remove every 2nd or 3rd row.

d) Remove Blank Rows

- Automatically deletes fully blank rows.

e) Remove Duplicates

- Keeps only unique rows based on selected columns.

Steps

Home → **Remove Rows** → select option.

Geetanjali

4. **Add New Columns**

In **Power Query Editor**, adding new columns is essential for:

- Data transformation
- Data cleaning
- Feature engineering
- Preparing data for DAX & visualizations

The **Add Column** tab provides multiple ways to create new columns depending on your requirement.

1. Custom Column: A **Custom Column** is used to create formulas using Power Query's M-language.

- Creating calculated fields
- Mathematical operations
- Text manipulation
- Date calculations
- Logical operations

Example

Formula to calculate **GST = 18% of Sales**:

=[Sales] * 0.18

More Examples

a) Concatenate Text

=[FirstName] & " " & [LastName]

b) Check Even/Odd

= if Number.Mod([ID], 2) = 0 then "Even" else "Odd"

c) Calculate Profit

=[Revenue] - [Cost]

2. Index Column

Adds a **sequential number**:
0,1,2… or 1,2,3…

Steps

Geetanjali

Add Column → **Index Column**

Options

- From 0
- From 1
- Custom starting number

It Is Used

- To identify the **order of rows**
- To remove duplicates and keep the first/last row
- To create row numbers
- For sorting in visuals or modeling
- To group data and retain sequence

Example

Row	Name
1	A
2	B

3. Conditional Column

What It Is

Creates a new column based on IF…ELSE logic **without coding**.

Example

If Sales > 5000 → "High"
Else → "Low"

Best Use Cases

- Classification (High/Low, Yes/No, Pass/Fail)
- Flagging missing or incorrect values
- Creating categories (Age group, Risk level)
- Grouping sales or performance ranges

4. Duplicate Column

Creates a **copy of an existing column**.

When It Is Needed

- You want to transform a column but keep the original

Geetanjali

- Preparing multiple variations of same field
- Converting text → uppercase without losing original
- Splitting a column but keeping original

Steps

Add Column → **Duplicate Column**

Example

If you duplicate "Name" column and transform to uppercase:

Name	Name - Copy
Geeta	GEETA

6. Other Add Column Options

Power Query provides many built-in column types.

a) Column from Examples

- Type sample output → Power BI generates logic
- Great for: extracting substring, formatting text, combining data

Example:
Input: "ABC-2024"
Output: "2024"

Power Query learns pattern → extracts year.

b) Date & Time Columns

You can add new columns from dates:

- Year
- Month
- Quarter
- Day
- Week number
- Start of month
- End of month

Example:
OrderDate → extract Year:

2025-04-18 → 2025

c) Extract & Format Columns

Geetanjali

- Extract prefix/suffix
- Extract first N characters
- Format numeric or text fields

Steps to Add Any New Column

General Steps

1. Open **Power Query Editor**
2. Go to **Add Column** tab
3. Choose the type:
    - **Custom Column**
    - **Index Column**
    - **Conditional Column**
    - **Duplicate Column**
    - **Invoke Function**
    - **Column From Examples**
    - **Math / Date / Time columns**
4. Enter details → Click **OK**

Geetanjali

4-Conditional Column

A **Conditional Column** allows you to add a new column based on **logical rules** (IF…ELSE statements) **without writing M-code**.

It is one of the most commonly used tools in Power Query for data cleaning, data transformation, and classification.

1. How to Add a Conditional Column

Steps

1. Go to **Power Query Editor**
2. Click **Add Column** tab
3. Select **Conditional Column**

You will see a dialog box where you can define:

- Column Name
- Column to evaluate
- Condition (operator)
- Value
- Output
- Else output

2. Example Scenario (Basic IF Condition)

Condition

If Sales > 5000 → "High"
Else → "Low"

Steps

1. **Name of new column:** *Sales Category*
2. **Column Name:** Sales
3. **Operator:** is greater than
4. **Value:** 5000
5. **Output:** High
6. **Else:** Low

Result Table

Sales	Sales Category
7000	High
4000	Low

Why Conditional Column is Useful

Geetanjali

No coding needed

You can build IF statements visually.

Perfect for classification tasks

- Categorizing sales (High/Medium/Low)
- Grading students
- Risk levels (Low/High)
- Profit evaluation (Profit/Loss)

Works well for Data Cleaning

- Replace values based on condition
- Flag missing or incorrect entries
- Identify outliers

Helps in Feature Engineering

Multiple Conditions (Nested IF)

You can add multiple conditions by clicking **Add Clause**.

Example: Categorizing Sales

Rule:

- If Sales > 8000 → "Excellent"
- If Sales > 5000 → "Good"
- Else → "Average"

Setup in Conditional Column

1. Column Name: Sales

Add clauses:

Condition	Value	Output
is greater than	8000	Excellent
is greater than	5000	Good
Else	—	Average

Result Example

Sales	Category
9000	Excellent
5600	Good

Geetanjali

Sales	Category
3000	Average

Conditional Column Operators

Comparison Operators

- equals
- does not equal
- is greater than
- is greater than or equal to
- is less than
- is less than or equal to

Text Operators

- begins with
- ends with
- contains
- does not contain

Date Operators

- is after
- is before
- is on or after
- is on or before

These help when creating conditions based on:

- Categories
- Dates
- Numeric values
- Substrings

Conditional Column Examples (Practical Use Cases)

a) Flagging Missing Values

If CustomerName is null → "Missing"
Else → "Valid"

b) Profit or Loss

If Profit >= 0 → "Profit"
Else → "Loss"

c) Age Grouping

Geetanjali

If Age < 18 → Child
If Age < 60 → Adult
Else → Senior

d) Date-Based Classification

If OrderDate is before 01-01-2024 → “Old Orders”

e) Grading Example

Marks: <40 = Fail, 40–60 = Pass, >60 = Merit

Geetanjali

Removing Blank / Null Values

Nulls and blanks are extremely common in imported datasets (Excel, CSV, SQL).
They create problems such as:

- Wrong totals (SUM ignores nulls but COUNT includes blanks)
- Relationship issues in data modeling
- Errors in DAX calculations
- Incorrect visuals (empty bars or missing categories)
- Performance slowdown during refresh

Power Query offers multiple ways to remove or handle them depending on the situation.

1. Remove Blank Rows (Full Row Null or Empty)

- When the entire row is empty
- When the row has no value in any column
- When blank rows appear due to formatting in Excel files

Steps

Home → **Remove Rows → Remove Blank Rows**

- Power Query checks each row.
- If **every column** in that row = null or " " → row gets deleted.

Example

Name	Age	Salary
Riya	26	30000
Amit	32	50000

After removal:

Name	Age	Salary
Riya	26	30000
Amit	32	50000

2. Remove Nulls from a Specific Column

When to Use

- When only **specific columns** cannot contain nulls
- For key columns like ID, ProductName, Date
- When dealing with relationship keys (Primary/Foreign Key)

Method 1 → Filter Method

Geetanjali

1. Click the **filter dropdown** in the column
2. Uncheck **(null)**
3. OK

Method 2 → Remove Blank Values Command

Home → Remove Rows → **Remove Blank Rows**
(This works only if the entire row is blank — not column-wise)

Example

Column "Employee_ID" contains:

Employee_ID
102
null
103

After filtering null:

Employee_ID
102
103

3. Replace Null Values (Using Replace Values)

- When you cannot remove rows
- When null needs a meaningful default value
- To avoid DAX errors (division by zero, type mismatch)
- To standardize data (e.g., replace null salaries with 0)

Steps

Home → **Replace Values**

Enter:

- Value to Find → **null**
- Replace With → **desired value**

Common Use Cases

Situation	Replacement
Missing numeric value	0
Missing category	"Unknown" or "Not Provided"
Missing date	Default date (like 01-Jan-2020)

Geetanjali

Situation	Replacement
Missing Boolean	False

Example

Salary column:

Salary
40000
null
50000

Replace null → 0:

Salary
40000
0
50000

Filter Out Nulls (Using Filters)

- When you want to **keep rest of the rows**, excluding the null ones
- Fastest method to remove unwanted null values
- When you want to quickly clean only one column

Steps

1. Select the **column filter** dropdown
2. Uncheck **(null)**
3. Apply filter

Example

"Department" column:

Department
HR
null
IT
Finance

After filter:

Geetanjali

Department
HR
IT
Finance

Best For

- Removing null category
- Cleaning duplicates with missing values
- Fixing categorical data before visuals

Remove Rows with Any Nulls Across Multiple Columns

Sometimes you need to remove rows where **any** important field is null.

Steps

1. Select multiple columns
2. Right-click → **Remove Empty**
   OR
   Use Custom Filtering

When Useful

- Data quality enforcement
- Ensuring complete records for modeling
- Preparing training data for ML models

6. Handling Nulls in Numeric Columns

Options

Method	Best Use
Replace with 0	Totals, calculations, missing numeric values
Remove	If the record is incomplete
Replace with Average/Median	Data normalization
Replace with Previous/Next value	Time-series data

Handling Nulls in Text Columns

Options

Method	Best Use
Replace with “Unknown”	Category fields

Geetanjali

Method	Best Use
Replace with “N/A”	Missing description
Remove	Duplicate or incomplete entries

Example:
“City” column: null → “Unknown”

Why Removing/Handling Null Values Matters

1. Prevents Wrong Calculations

- SUM results less
- COUNT becomes more
- Averages become inaccurate

2. Prevents Relationship Errors

Power BI relationships break when:

- Key column has null
- Foreign key has null

3. Improves Model Quality

Clean data = faster refresh, accurate measures.

4. Avoids Visual Problems

- Empty bars
- Missing categories
- Wrong filtering

5. Essential for Data Modeling

Null values reduce data integrity and cause misinterpretation.

Geetanjali